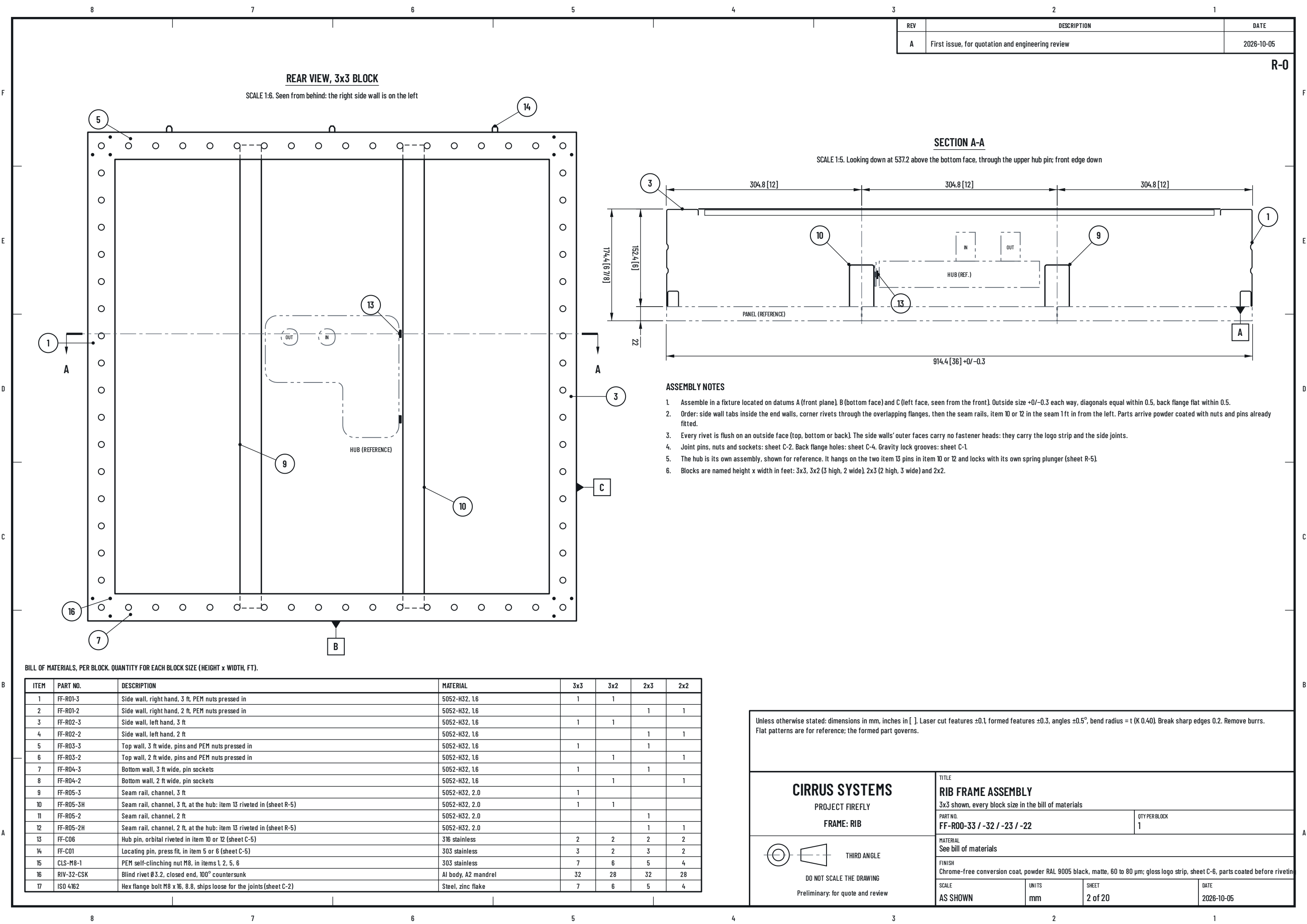




B

BILL OF MATERIALS, PER BLOCK. QUANTITY FOR EACH BLOCK SIZE (HEIGHT x WIDTH, FT).

ITEM	PART NO.	DESCRIPTION	MATERIAL	3x3	3x2	2x3	2x2
1	FF-R01-3	Side wall, right hand, 3 ft, PEM nuts pressed in	5052-H32, 1.6	1	1		
2	FF-R01-2	Side wall, right hand, 2 ft, PEM nuts pressed in	5052-H32, 1.6			1	1
3	FF-R02-3	Side wall, left hand, 3 ft	5052-H32, 1.6	1	1		
4	FF-R02-2	Side wall, left hand, 2 ft	5052-H32, 1.6			1	1
5	FF-R03-3	Top wall, 3 ft wide, pins and PEM nuts pressed in	5052-H32, 1.6	1		1	
6	FF-R03-2	Top wall, 2 ft wide, pins and PEM nuts pressed in	5052-H32, 1.6		1		1
7	FF-R04-3	Bottom wall, 3 ft wide, pin sockets	5052-H32, 1.6	1		1	
8	FF-R04-2	Bottom wall, 2 ft wide, pin sockets	5052-H32, 1.6		1		1
9	FF-R05-3	Seam rail, channel, 3 ft	5052-H32, 2.0	1			
10	FF-R05-3H	Seam rail, channel, 3 ft, at the hub: item 13 riveted in (sheet R-5)	5052-H32, 2.0	1	1		
11	FF-R05-2	Seam rail, channel, 2 ft	5052-H32, 2.0			1	
12	FF-R05-2H	Seam rail, channel, 2 ft, at the hub: item 13 riveted in (sheet R-5)	5052-H32, 2.0			1	1
13	FF-C06	Hub pin, orbital riveted in item 10 or 12 (sheet C-5)	316 stainless	2	2	2	2
14	FF-C01	Locating pin, press fit, in item 5 or 6 (sheet C-5)	303 stainless	3	2	3	2
15	CLS-M8-1	PEM self-clinching nut M8, in items 1, 2, 5, 6	303 stainless	7	6	5	4
16	RIV-32-CSK	Blind rivet Ø3.2, closed end, 100° countersunk	Al body, A2 mandrel	32	28	32	28
17	ISO 4162	Hex flange bolt M8 x 16, 8.8, ships loose for the joints (sheet C-2)	Steel, zinc flake	7	6	5	4

CIRRUS SYSTEMS

PROJECT FIREFLY

FRAME: RIB

THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

RIB FRAME ASSEMBLY

3x3 shown, every block size in the bill of materials

PART NO.

FF-R00-33 / -32 / -23 / -22

QTY PER BLOCK

1

MATERIAL

See bill of materials

FINISH

Chrome-free conversion coat, powder RAL 9005 black, matte, 60 to 80 µm; gloss logo strip, sheet C-6, parts coated before riveting

SCALE

AS SHOWN

UNITS

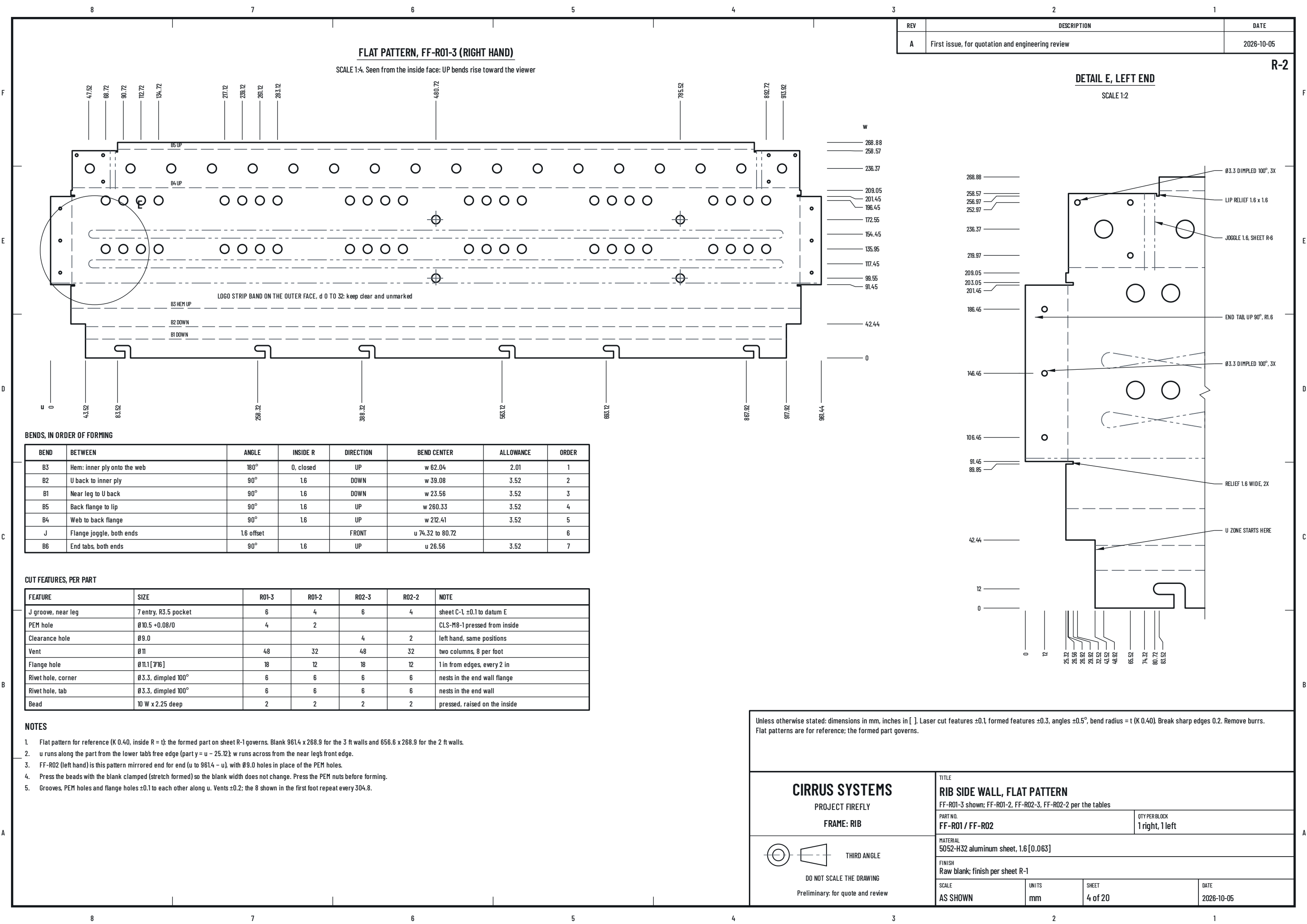
mm

SHEET

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DATE

2026-10-05



NOTES

- Flat pattern for reference (K 0.40, inside R = t): the formed part on sheet R-1 governs. Blank 961.4 x 268.9 for the 3 ft walls and 656.6 x 268.9 for the 2 ft walls.
- u runs along the part from the lower tabs free edge (part y = u - 25.12); w runs across from the near legs front edge.
- FF-R02 (left hand) is this pattern mirrored end for end (u to 961.4 - u), with Ø9.0 holes in place of the PEM holes.
- Press the beads with the blank clamped (stretch formed) so the blank width does not change. Press the PEM nuts before forming.
- Grooves, PEM holes and flange holes ±0.1 to each other along u. Vents ±0.2; the 8 shown in the first foot repeat every 304.8.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ±0.1, formed features ±0.3, angles ±0.5°, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

FRAME: RIB



THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

RIB SIDE WALL, FLAT PATTERN

FF-R01-3 shown; FF-R01-2, FF-R02-3, FF-R02-2 per the tables

PART NO.

FF-R01 / FF-R02

QTY PER BLOCK

1 right, 1 left

MATERIAL

5052-H32 aluminum sheet, 1.6 [0.063]

FINISH

Raw blank; finish per sheet R-1

SCALE

AS SHOWN

UNITS

mm

SHEET

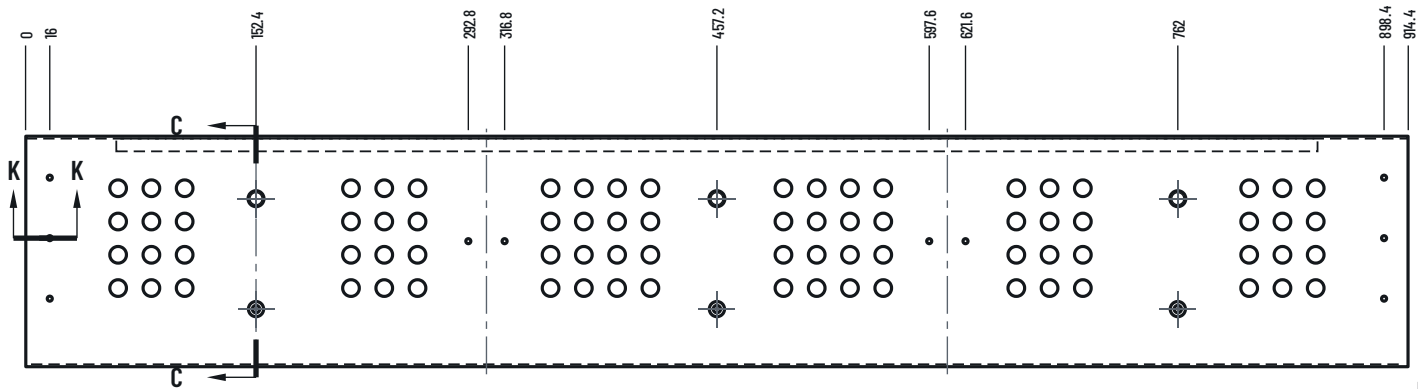
4 of 20

DATE

2026-10-05

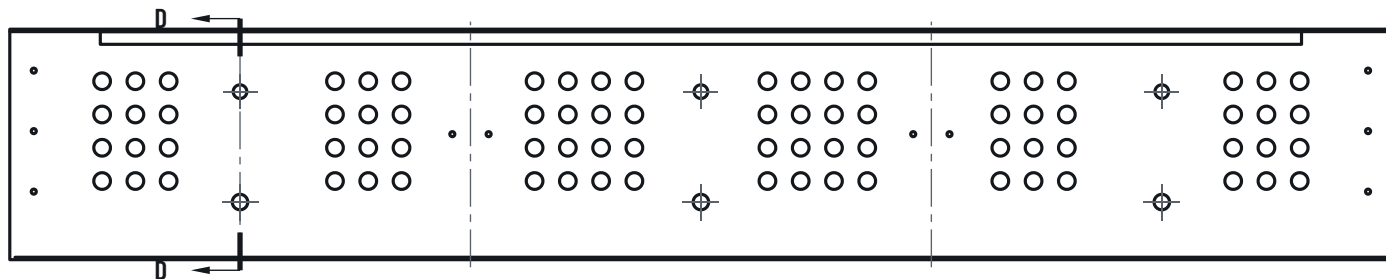
REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05

R-3



TOP WALL FF-R03-3, SEEN FROM ABOVE

SCALE 1:5. x from datum C, the block's left face; front edge down

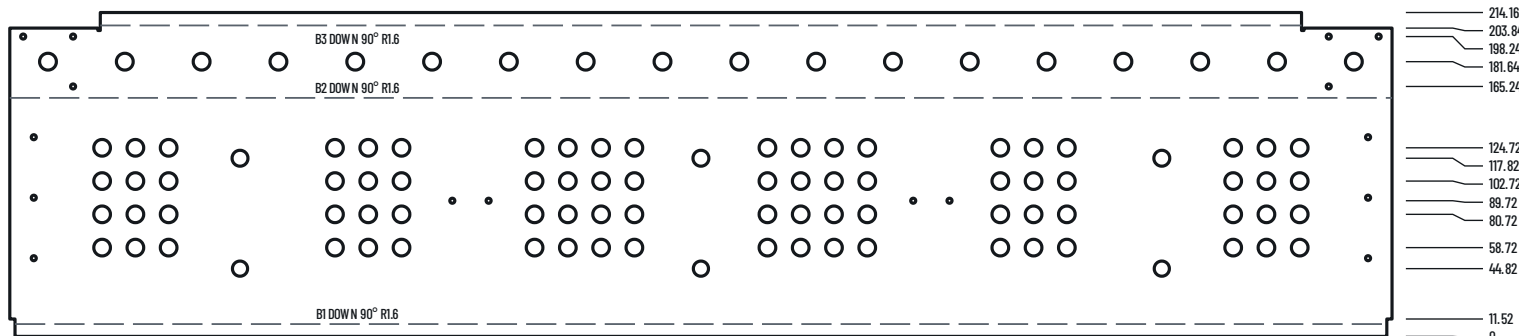


BOTTOM WALL FF-R04-3, SEEN FROM ABOVE (INSIDE)

SCALE 1:5. Same x positions as the top wall

FLAT PATTERN, FF-R03-3

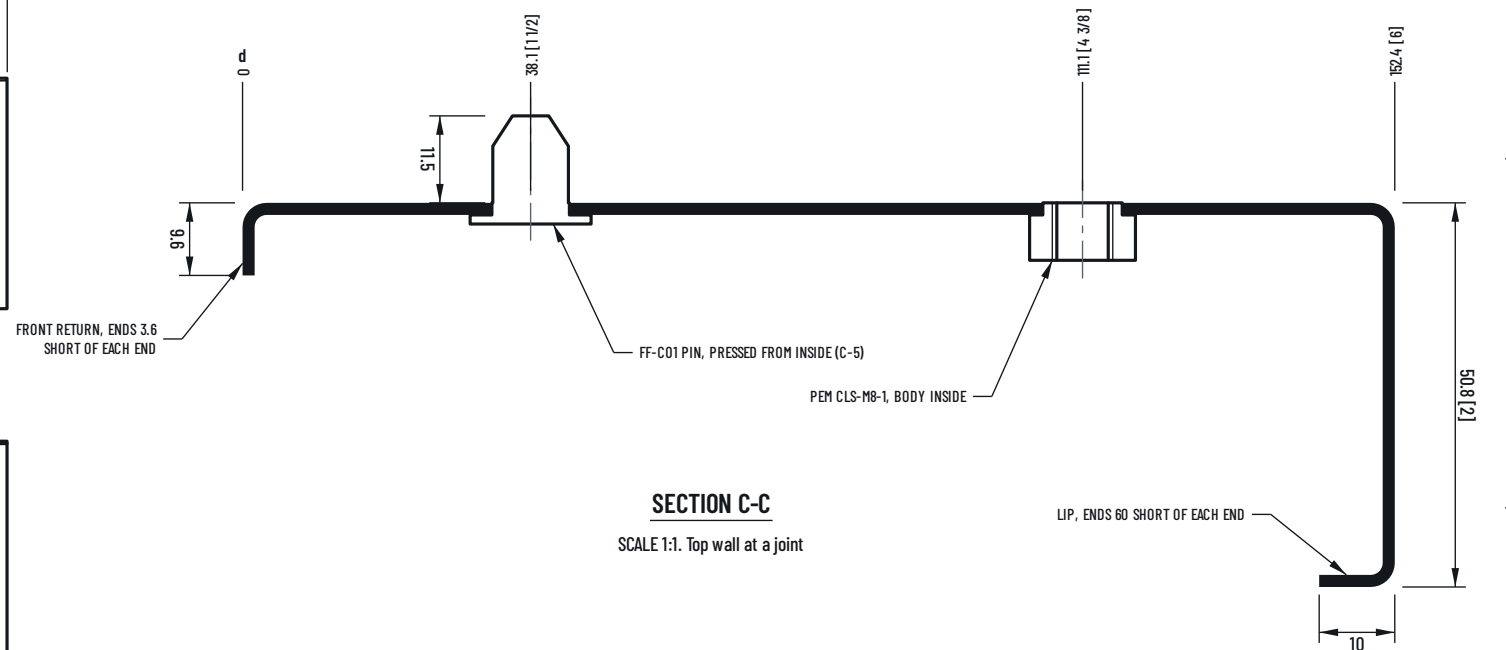
SCALE 1:5. Seen from the outside face; every bend goes away from the viewer



PART NO.	WALL	LENGTH	FRONT LINE, d 38.1	BACK LINE, d 11.1	VENTS	SEAM RIVET HOLES	BLANK
FF-R03-3	Top	914.1	3 pins FF-C01 in Ø10.0	3 PEM CLS-M8-1 in Ø10.5	80	4	914.1 x 214.2
FF-R03-2	Top	609.3	2 pins FF-C01 in Ø10.0	2 PEM CLS-M8-1 in Ø10.5	56	2	609.3 x 214.2
FF-R04-3	Bottom	914.1	3 sockets Ø10.2 +0.1/0	3 holes Ø9.0	80	4	914.1 x 214.2
FF-R04-2	Bottom	609.3	2 sockets Ø10.2 +0.1/0	2 holes Ø9.0	56	2	609.3 x 214.2

NOTES

1. Length = block width - 0.3 ±0.1 so an end wall is never proud of the side faces. Pins, sockets, nuts and holes at every foot center, $x = 152.4 [6] + 304.8 [12] n$.
2. Vents Ø11 in four rows (d 52, 74, 96, 118) at 22 pitch, centered, clear of every foot center by 36 and every seam rail by 20 (hole edge to zone). 3 ft: x 61.2 to 105.2, 215.2 to 259.2, 347.2 to 413.2, 501.2 to 567.2, 655.2 to 699.2, 809.2 to 853.2.
3. Press the pins and nuts from the inside before forming; mask pin shanks, sockets and nut threads for powder coat. Pins and sockets ±0.05 to each other, ±0.1 to datum C.

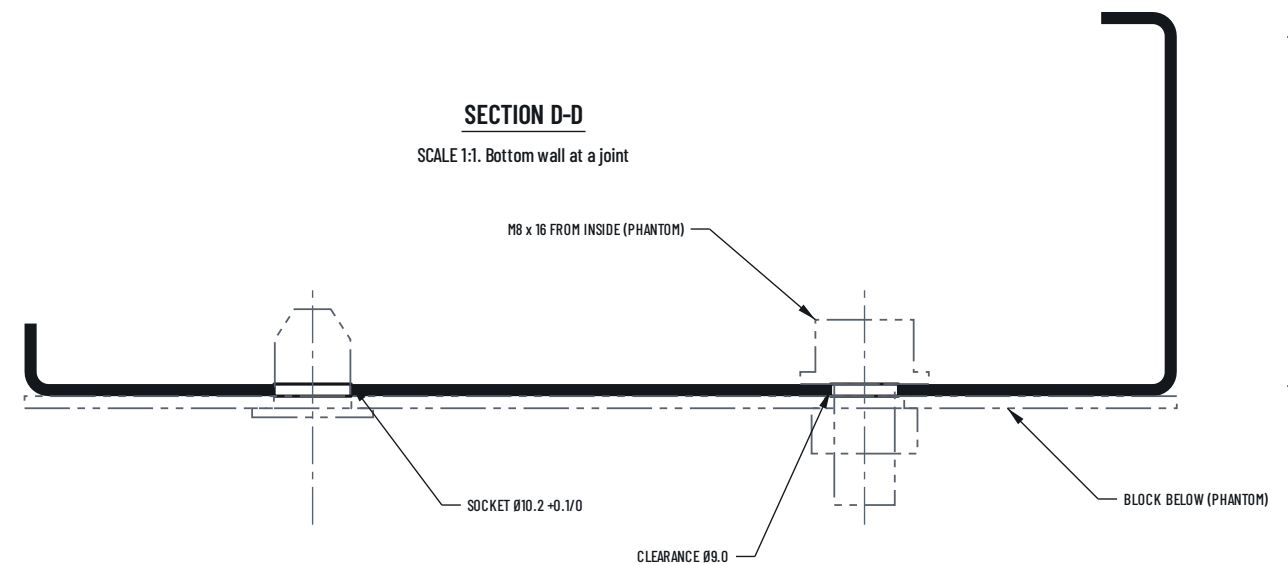


SECTION C-C

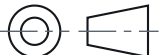
SCALE 1:1. Top wall at a joint

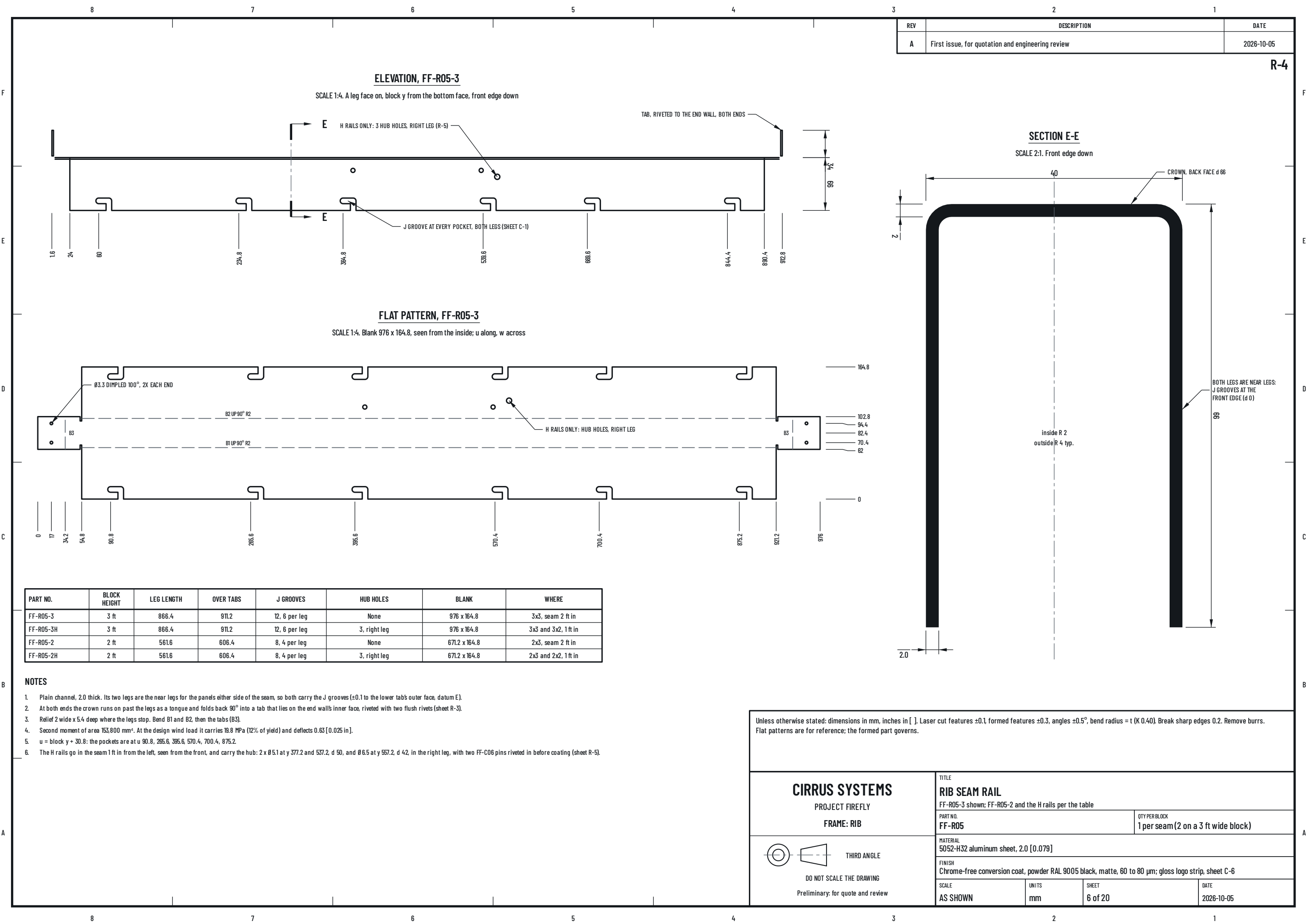
SECTION D-D

SCALE 1:1. Bottom wall at a joint



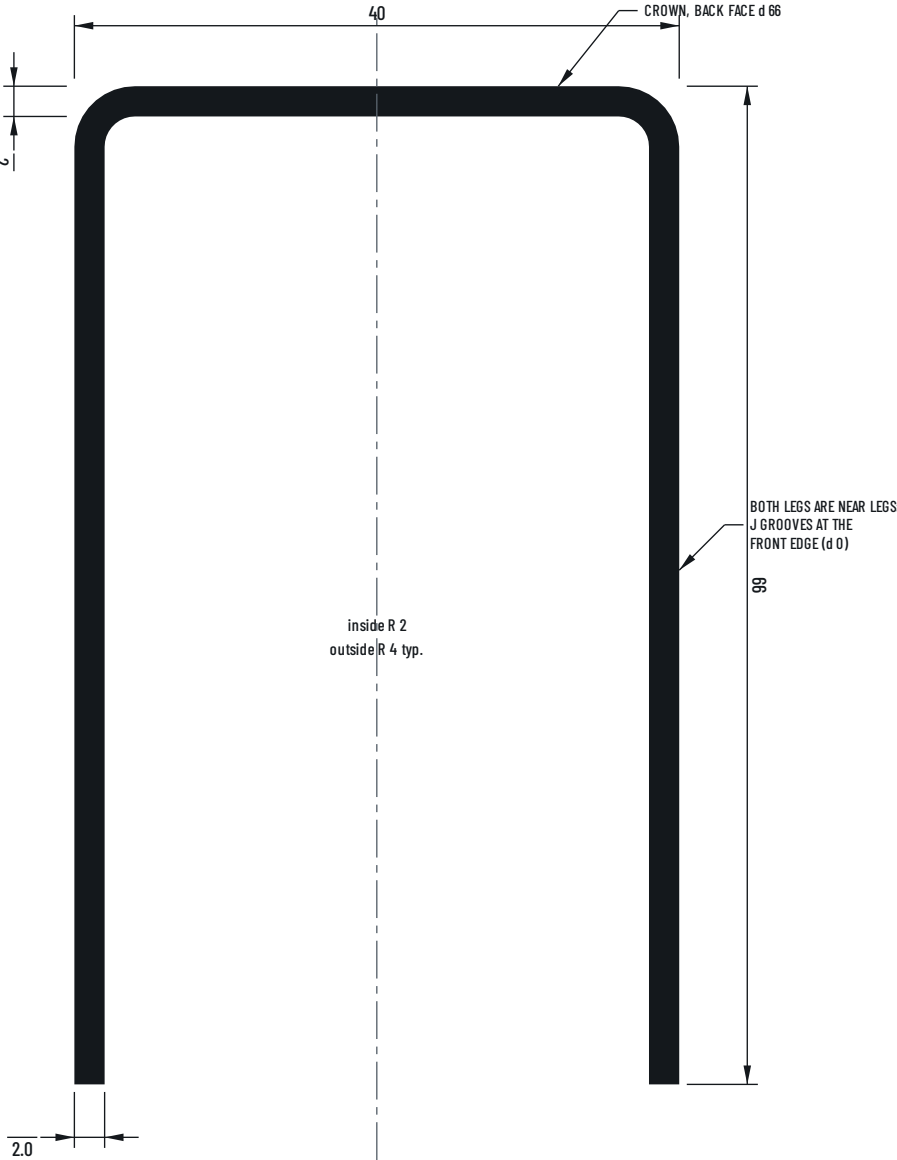
Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

 THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	TITLE CIRRUS SYSTEMS PROJECT FIREFLY FRAME: RIB		
	RIB TOP AND BOTTOM WALLS FF-R03-3, FF-R03-2 (top), FF-R04-3, FF-R04-2 (bottom)		
	PART NO. FF-R03 / FF-R04	QTY PER BLOCK 1 top, 1 bottom	
	MATERIAL 5052-H32 aluminum sheet, 1.6 [0.063] FINISH Chrome-free conversion coat, powder RAL 9005 black, matte, 60 to 80 µm; gloss logo strip, sheet C-6		
SCALE AS SHOWN	UNITS mm	SHEET 5 of 20	DATE 2026-10-05



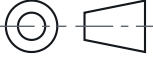
REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05

R-4



PART NO.	BLOCK HEIGHT	LEG LENGTH	OVER TABS	J GROOVES	HUB HOLES	BLANK	WHERE
FF-R05-3	3 ft	866.4	911.2	12, 6 per leg	None	976 x 164.8	3x3, seam 2 ft in
FF-R05-3H	3 ft	866.4	911.2	12, 6 per leg	3, right leg	976 x 164.8	3x3 and 3x2, 1 ft in
FF-R05-2	2 ft	561.6	606.4	8, 4 per leg	None	671.2 x 164.8	2x3, seam 2 ft in
FF-R05-2H	2 ft	561.6	606.4	8, 4 per leg	3, right leg	671.2 x 164.8	2x3 and 2x2, 1 ft in

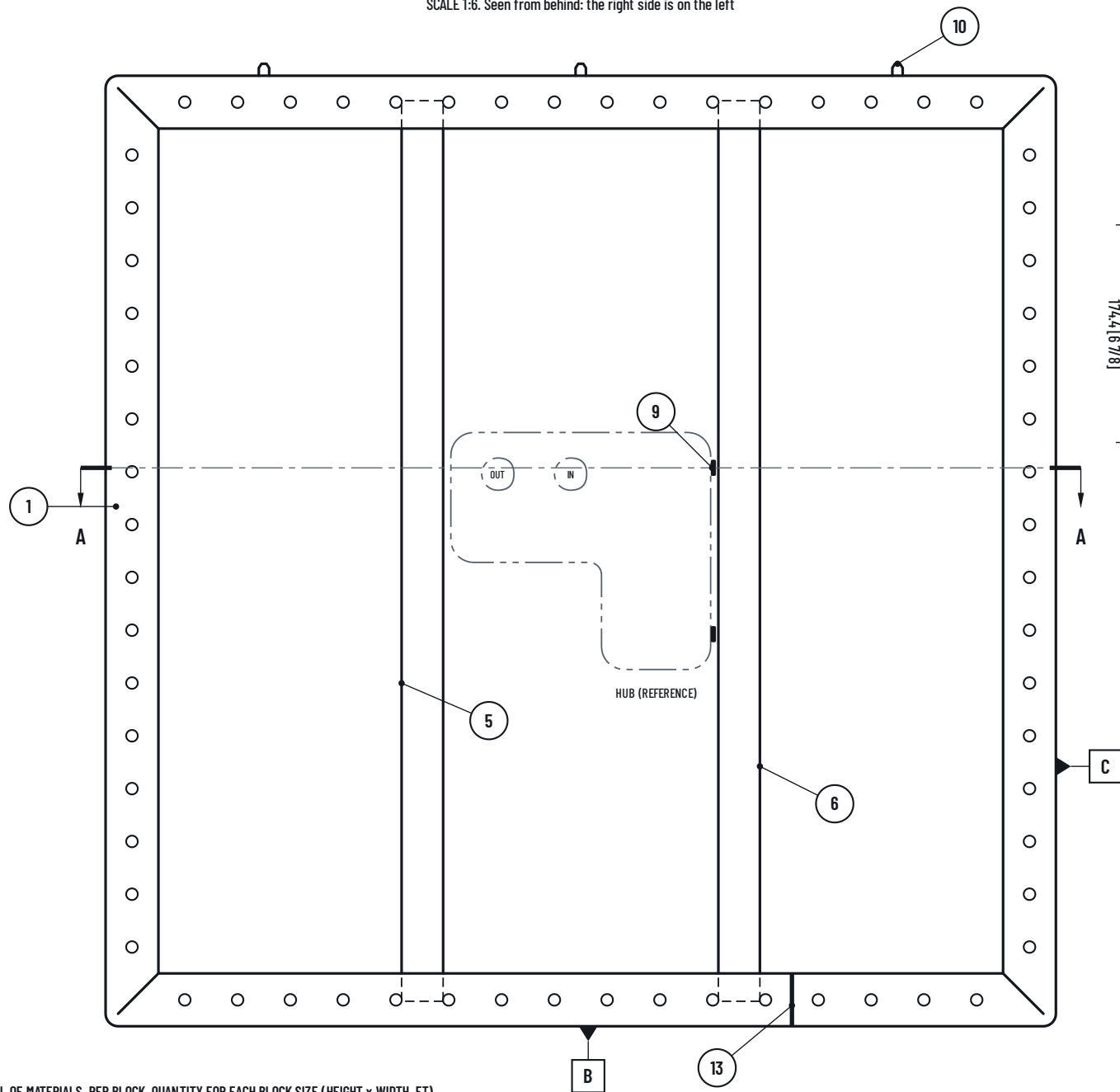
- NOTES**
- Plain channel, 2.0 thick. Its two legs are the near legs for the panels either side of the seam, so both carry the J grooves (± 0.1 to the lower tab's outer face, datum E).
 - At both ends the crown runs on past the legs as a tongue and folds back 90° into a tab that lies on the end wall's inner face, riveted with two flush rivets (sheet R-3).
 - Relief 2 wide x 5.4 deep where the legs stop. Bend B1 and B2, then the tabs (B3).
 - Second moment of area 153,800 mm⁴. At the design wind load it carries 19.8 MPa (12% of yield) and deflects 0.63 [0.025 in].
 - u = block y + 30.8: the pockets are at u 90.8, 285.6, 395.6, 570.4, 700.4, 875.2
 - The H rails go in the seam 1 ft in from the left, seen from the front, and carry the hub: 2 x $\varnothing 5.1$ at y 377.2 and 537.2, d 50, and $\varnothing 6.5$ at y 557.2, d 42, in the right leg, with two FF-C06 pins riveted in before coating (sheet R-5).

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.			
CIRRUS SYSTEMS PROJECT FIREFLY FRAME: RIB		TITLE RIB SEAM RAIL FF-R05-3 shown; FF-R05-2 and the H rails per the table	
		PART NO. FF-R05	QTY PER BLOCK 1 per seam (2 on a 3 ft wide block)
 THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review		MATERIAL 5052-H32 aluminum sheet, 2.0 [0.079]	
		FINISH Chrome-free conversion coat, powder RAL 9005 black, matte, 60 to 80 μ m; gloss logo strip, sheet C-6	
		SCALE AS SHOWN	UNITS mm
		SHEET 6 of 20	DATE 2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05

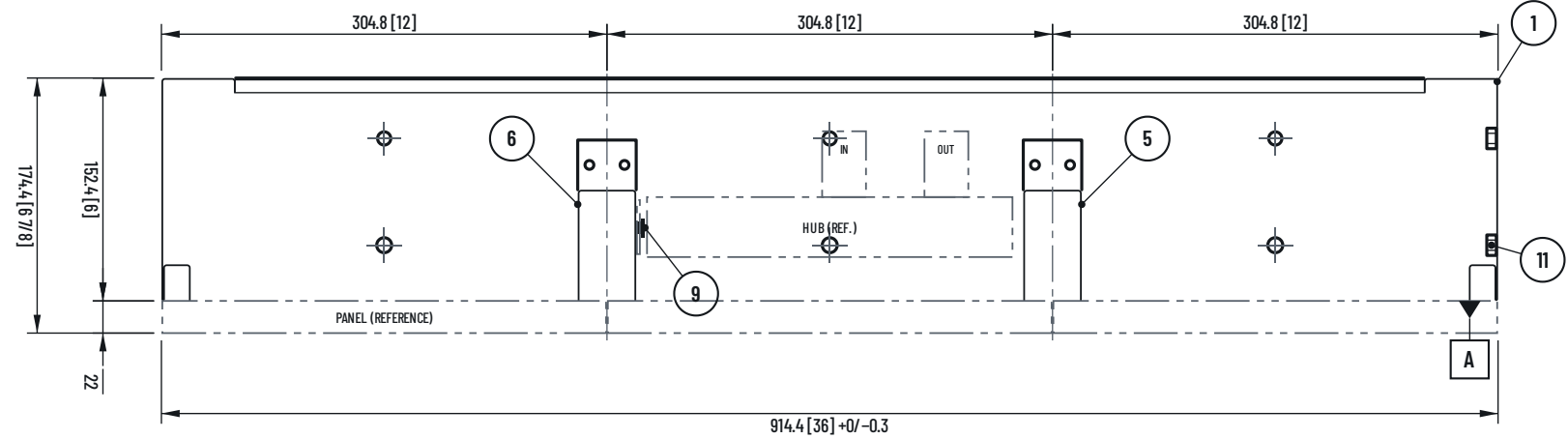
REAR VIEW, 3x3 BLOCK

SCALE 1:6. Seen from behind: the right side is on the left



SECTION A-A

SCALE 1:5. Looking down at 537.2 above the bottom face, through the upper hub pin; front edge down. Vents and flange holes not shown



ASSEMBLY NOTES

1. Fixture on datums A (front plane), B (bottom face) and C (left face, seen from the front). Outside size $+0/-0.3$ each way, diagonals equal within 1.0, back flange flat within 0.5.
2. Order: close the ring with its one weld (sheet H-1), projection weld the nuts and locating pins, clinch in the seam rails (item 6 or 8, its hub pins already riveted, in the seam 1 ft in from the left), mask the hub pins, locating pins, sockets and every nut thread, then clean, pretreat and powder coat the whole frame matte black. Pad print the gloss logo strip up both sides (sheet C-6).
3. Nothing is fastened to a side wall's outer face: it carries the logo strip and the side joints. Clinch dishes sit on the top, bottom and back faces only, never proud.
4. Joint pins, nuts and sockets: sheet C-2. Back flange holes: sheet C-4 (no corner holes on Hoop). Gravity lock grooves: sheet C-1. Hub mount: sheet H-5.
5. The hub is its own assembly, shown for reference. It hangs on the two item 9 pins in item 6 or 8 and locks with its own spring plunger (sheet H-5).
6. Item 12 ships loose: one bolt for each clearance hole in the bottom wall and in the left side wall (seen from the front). Each bolts into the next block's weld nut, or into the base rail under the bottom row (sheets C-2 and C-3).
7. Blocks are named height x width in feet: 3x3, 3x2 (3 high, 2 wide), 2x3 (2 high, 3 wide) and 2x2.

EACH BLOCK SIZE

BLOCK	ASSEMBLY	OUTSIDE W x H x D	SEAM RAILS	STRIP, kg	FRAME, kg (APPROX.)
3x3	FF-H00-33	914.4 x 914.4 x 152.4	2	8.7	12.2
3x2	FF-H00-32	609.6 x 914.4 x 152.4	1	7.2	9
2x3	FF-H00-23	914.4 x 609.6 x 152.4	2	7.2	9.7
2x2	FF-H00-22	609.6 x 609.6 x 152.4	1	5.7	7

1. Frame mass includes the seam rails, nuts and pins, before powder coat; the panels and hub are not included.

BILL OF MATERIALS, PER BLOCK. QUANTITY FOR EACH BLOCK SIZE (HEIGHT x WIDTH, FT).

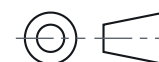
ITEM	PART NO.	DESCRIPTION	MATERIAL	3x3	3x2	2x3	2x2
1	FF-H01-33	Hoop strip for a 3x3, punched, formed, bent and welded	S280GD+ZM310, 1.2	1			
2	FF-H01-32	Hoop strip for a 3x2	S280GD+ZM310, 1.2		1		
3	FF-H01-23	Hoop strip for a 2x3	S280GD+ZM310, 1.2			1	
4	FF-H01-22	Hoop strip for a 2x2	S280GD+ZM310, 1.2				1
5	FF-H02-3	Seam rail, channel, 3 ft, clinched (sheet H-4)	S280GD+ZM310, 1.2	1			
6	FF-H02-3H	Seam rail, channel, 3 ft, at the hub: item 9 riveted in (sheet H-5)	S280GD+ZM310, 1.2	1	1		
7	FF-H02-2	Seam rail, channel, 2 ft, clinched	S280GD+ZM310, 1.2			1	
8	FF-H02-2H	Seam rail, channel, 2 ft, at the hub: item 9 riveted in (sheet H-5)	S280GD+ZM310, 1.2			1	1
9	FF-C06	Hub pin, orbital riveted in item 6 or 8 (sheet C-5)	316 stainless	2	2	2	2
10	FF-C02	Locating pin, projection welded in the top wall (sheet C-5)	Steel, zinc nickel	3	2	3	2
11	DIN 929	Hex weld nut M8, top wall and right side wall	Steel, plain	7	6	5	4
12	ISO 4162	Hex flange bolt M8 x 16, 8.8, ships loose for the joints (sheet C-2)	Steel, zinc flake	7	6	5	4
13	WELD	Square butt, one per ring, x 254 on the bottom (sheet H-1)		1	1	1	1

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

FRAME: HOOP



THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

	TITLE
--	-------

HOOP FRAME ASSEMBLY

3x3 shown, every block size in the bill of materials

PART NO.

FF-H00-33 / -32 / -23 / -22

QTY PER BLOCK

1

	MATERIAL
--	----------

See bill of materials

FINISH

Zirconium pretreat, powder RAL 9005 black, matte, 60 to 80 μm ; gloss logo strip, sheet C-6, after welding and clinching

SCALE

AS SHOWN

	UNITS
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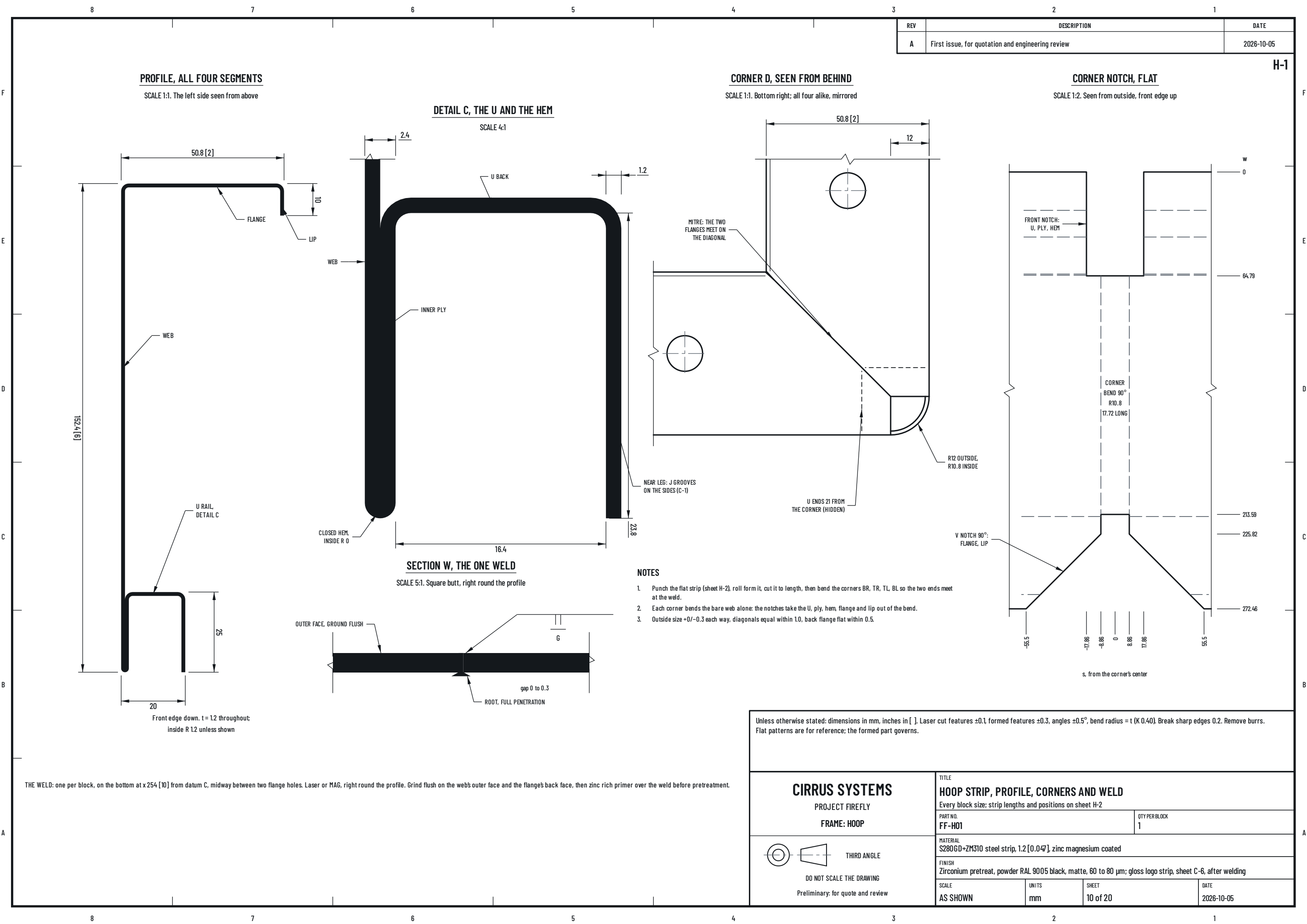
mm

SHEET

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DATE _____

2026-10-05

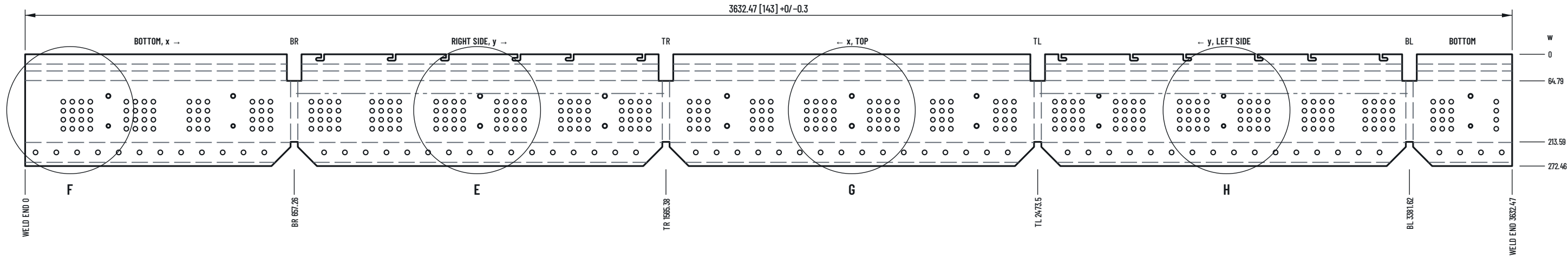


REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05

FLAT STRIP, FF-H01-33 (3x3 BLOCK)

SCALE 1:10. Seen from outside, front edge up, s from the weld end to the right

H-2



Details E to H on sheet H-3; F shows both ends butted at the weld. Corner bends run 8.86 either side of each center.

STRIP FOR EACH BLOCK SIZE (HEIGHT x WIDTH, FT)

PART NO.	BLOCK	OUTSIDE W x H	LENGTH L	s, BR	s, TR	s, TL	s, BL	MASS, kg
FF-H01-33	3x3	914.4 x 914.4	3632.47	657.26	1565.38	2473.5	3381.62	8.7
FF-H01-32	3x2	609.6 x 914.4	3022.87	352.46	1260.58	1863.9	2772.02	7.2
FF-H01-23	2x3	914.4 x 609.6	3022.87	657.26	1260.58	2168.7	2772.02	7.2
FF-H01-22	2x2	609.6 x 609.6	2413.27	352.46	955.78	1559.1	2162.42	5.7

CUT FEATURES AND HARDWARE PER STRIP

PART NO.	JOINT HOLES	VENTS	FLANGE HOLES	J GROOVES	CORNER NOTCHES	WELD NUTS DIN 929	PINS FF-C02
FF-H01-33	20	344	64	12	8	7	3
FF-H01-32	16	296	52	12	8	6	2
FF-H01-23	16	280	52	8	8	5	3
FF-H01-22	12	232	40	8	8	4	2

EVERY CUT FEATURE, ALONG ITS SEGMENT IN BLOCK COORDINATES

FEATURE	SEGMENT	ACROSS, w	SIZE	ALONG, 3 FT SEGMENT	ALONG, 2 FT SEGMENT
Pin socket	Bottom	101.69 (d 38.1)	Ø10.2 +0.1/0	x 152.4, 457.2, 762	x 152.4, 457.2
Bolt clearance	Bottom	174.69 (d 111.1)	Ø9.0	x 152.4, 457.2, 762	x 152.4, 457.2
Pin FF-CO2 seat	Top	101.69 (d 38.1)	Ø10.2 +0.1/0	x 152.4, 457.2, 762	x 152.4, 457.2
Weld nut DIN 929 M8	Top	174.69 (d 111.1)	Ø10.6 +0.1/0	x 152.4, 457.2, 762	x 152.4, 457.2
Weld nut DIN 929 M8	Right side	101.69 (d 38.1), 174.69	Ø10.6 +0.1/0	y 457.2, 762	y 457.2
Bolt clearance	Left side	101.69 (d 38.1), 174.69	Ø9.0	y 457.2, 762	y 457.2
J groove, pocket	Both sides	0 to 15.5	7 wide, R3.5	y 60, 234.8, 364.8, 539.6, 669.6, 844.4	y 60, 234.8, 364.8, 539.6
Vent, 4 rows	Bottom	115.59, 137.59, 159.59, 181.59	Ø11, 22 pitch	x 61.2 (3), 215.2 (3), 347.2 (4), 501.2 (4), 655.2 (3), 809.2 (3)	x 51.8 (3), 205.8 (2), 337.8 (4), 513.8 (3)
Vent, 4 rows	Top	the same	Ø11, 22 pitch	x 61.2 (3), 215.2 (3), 347.2 (4), 501.2 (4), 655.2 (3), 809.2 (3)	x 51.8 (3), 205.8 (4), 337.8 (4), 513.8 (3)
Vent, 4 rows	Both sides	the same	Ø11, 22 pitch	y 45.2 (4), 193.6 (4), 350 (4), 498.4 (4), 654.8 (4), 803.2 (4)	y 45.2 (4), 193.6 (4), 350 (4), 498.4 (4)
Flange hole	All four	239.22 (q 25.4)	Ø11.1, 50.8 pitch	from 76.2 (16)	from 76.2 (10)

MANUFACTURING ROUTE, ONE BLOCK

STEP	OPERATION	TOOLING	SHEET
1	Slit coil to 272.46 wide; decoil, level and feed	Low	H-2
2	Punch every hole, groove and notch in register	High	H-2, H-3
3	Roll form the profile; the hem closes in the last passes	High	H-1
4	Cut to length L after forming	Medium	H-2
5	Bend the four corners in a fixture: BR, TR, TL, BL	Medium	H-1
6	Butt weld the ends; grind flush; prime the weld	Medium	H-1
7	Projection weld the nuts and pins from inside	Low	H-0
8	Rivet the hub pins into the H rail; clinch in the seam rails	Low	H-0, H-5
9	Mask pins, sockets and nut threads; clean, pretreat, powder coat	Low	H-0
10	Pad print the gloss logo strip up both sides	Low	C-6

1. x runs from the block's left outer face seen from the front; y up from its bottom outer face. On a flat, s = the run's start + the distance from its corner (table above and view A). A count in () is that many centers at the pitch, starting at the one given.
2. d is depth back from the frame's front edge, q is in from the outer face along the flange; w = d + 63.59 on the web and q + 213.82 on the flange. Positions ± 0.1 to the strips punched datum, accumulated ± 0.3 along the full length.

NOTES

1. Strip: 272.46 +0.2/0 wide, slit from coil, ZM310 both sides, burr toward the inside face.
2. Punch every feature in register before roll forming; the J grooves, joint holes and corner notches share one datum per block length. Cut to length after forming.
3. The weld closes the ring at x 254 on the bottom, between two flange holes and clear of vents (sheet H-1).
4. Logo strip band: up both sides the outer face from the front edge to the phantom line at w 95.59 (d 32) carries no cut feature, burr, tooling mark or scratch. The logos are printed after coating, sheet C-6.
5. Mass is the bare formed strip, before nuts, pins and coating.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

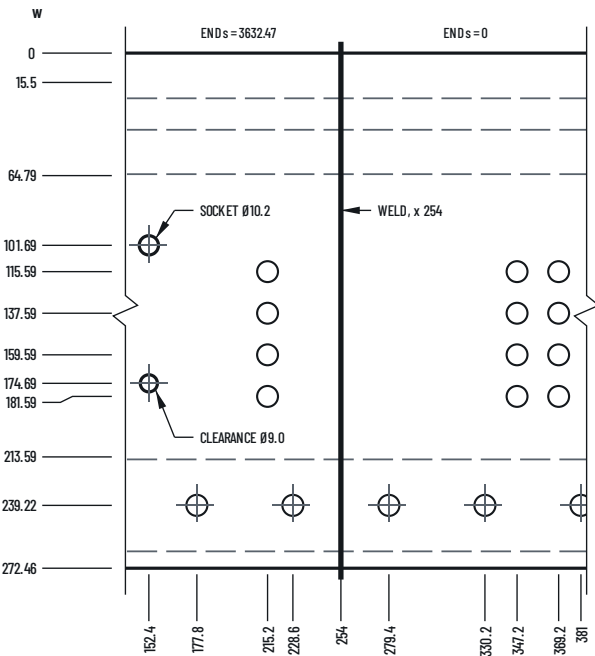
CIRRUS SYSTEMS PROJECT FIREFLY FRAME: HOOP	TITLE HOOP STRIP, FLAT PATTERN Strip lengths, corner positions and every cut feature for the four block sizes			
	PART NO. FF-H01-33 / -32 / -23 / -22		QTY PER BLOCK 1	
  THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	MATERIAL S280GD+ZM310 steel strip, 1.2 [0.047], zinc magnesium coated			
	FINISH Bare at this stage: coated after welding (sheet H-1)			
	SCALE 1:10	UNITS mm	SHEET 11 of 20	DATE 2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05

H-3

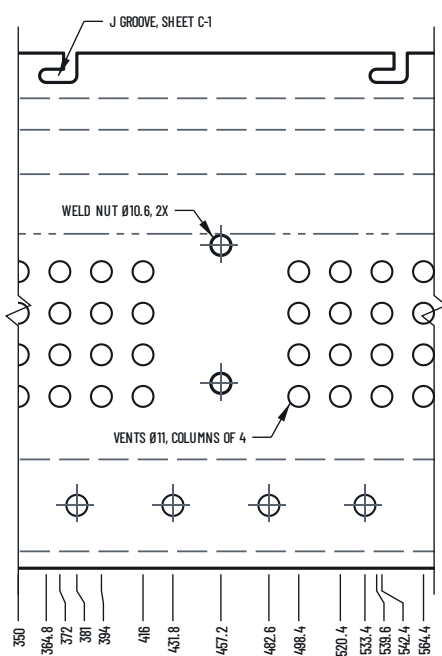
DETAIL F, BOTTOM

x → . Ends butted at the weld



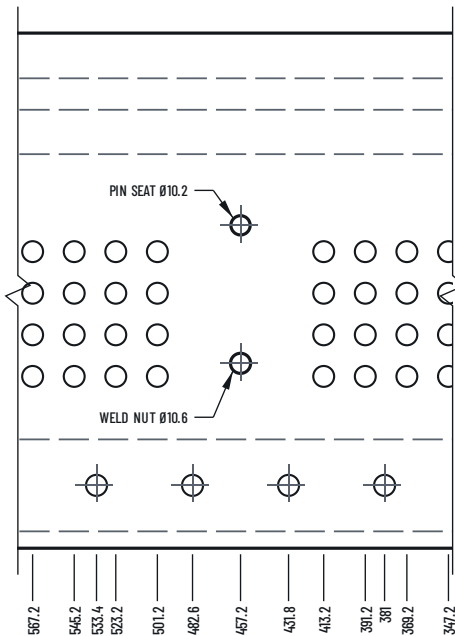
DETAIL E, RIGHT SIDE

$y \rightarrow .$ At a joint station



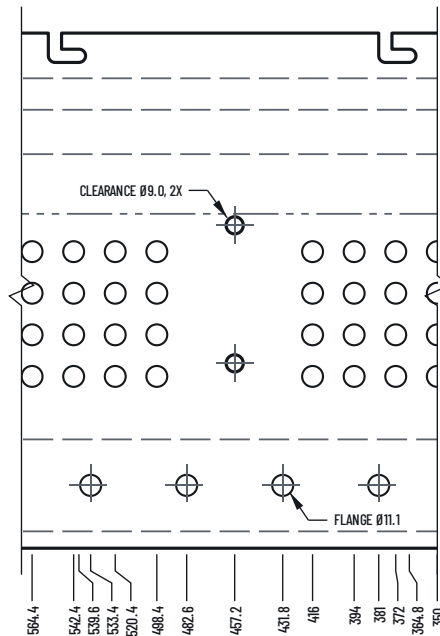
DETAIL G, TOP

← x. At a foot



DETAIL H, LEFT SIDE

← y. At a joint station



All four windows are on FF-H01-33, at 1:4 and at one height, so every row reads straight across. Positions under each window are block x or y.

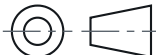
PUNCH TOOLS AND HITS, ONE FF-H01-33 STRIP

TOOL	SHAPE AND SIZE	MAKES	HITS	TOOLING
P1	Round Ø9.0	Bolt clearance, bottom and left	7	Low
P2	Round Ø10.2 +0.1/0	Pin sockets and pin seats	6	Low
P3	Round Ø10.6 +0.1/0	Weld nut pilots, top and right	7	Low
P4	Gang of 4, Ø11 at 22	One vent column per hit	86	Medium
P5	Round Ø11.1 [7/16]	Back flange holes	64	Low
P6	J groove, 7 wide, R3.5	Near leg, both sides	12	Medium
P7	Front notch 35.72 x 64.79	U, ply and hem out of each corner	4	Medium
P8	V notch 90°, III at the lip	Flange and lip mitre at each corner	4	Medium
P9	Cutoff, full profile	Length L, after forming	1	High

NOTES

1. Punch with the burr toward the inside face. Joint holes Ø9.0 to Ø10.6 true position 0.2 to the punched datum of the block length.
2. Vent columns never fall within 36 of a foot center, 20 of a seam rail or 15 of the weld; the patterns repeat each foot so vents line up block to block.
3. The phantom line in details E and H bounds the logo strip band, w 95.59 (d 32); nothing is punched and no tool marks the outer face between it and the front edge. The logos are printed after coating, sheet C-6.
4. Weld nut pilots and pin seats are sized for the parts on sheet C-5.

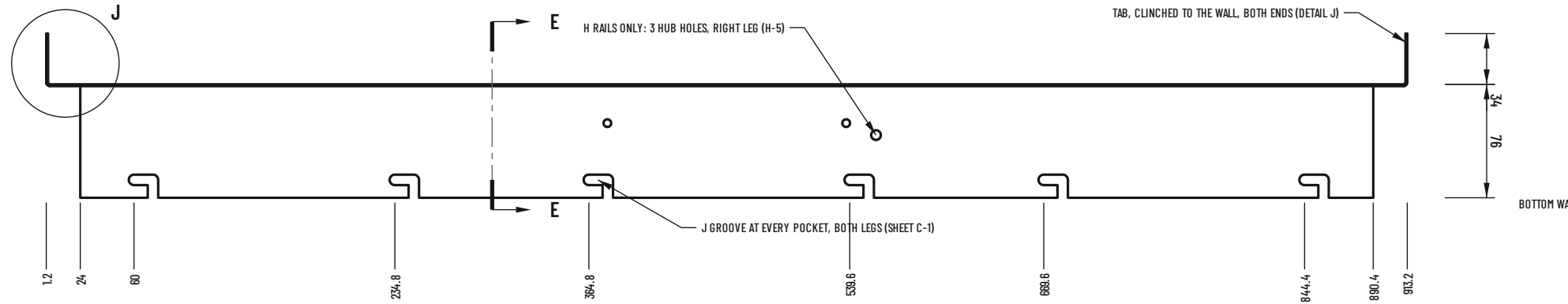
Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY FRAME: HOOP	TITLE		
	HOOP STRIP, PUNCH DETAILS		
 THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	Windows on FF-H01-33; every size uses the same tools		
	PART NO.		QTY PER BLOCK
	FF-H01-33 / -32 / -23 / -22		1
	MATERIAL		
S280GD+ZM310 steel strip, 1.2 [0.047], zinc magnesium coated			
FINISH			
Bare at this stage: coated after welding (sheet H-1)			
SCALE	UNITS	SHEET	DATE
AS SHOWN	mm	12 of 20	2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05

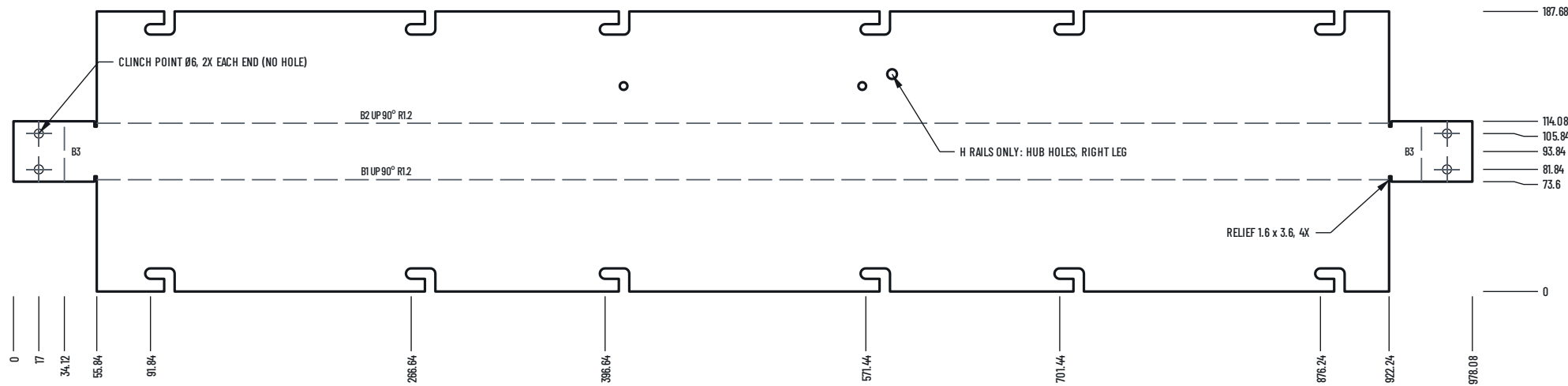
ELEVATION, FF-H02-3

SCALE 1:4. A leg face on, block y from the bottom face, front edge down



FLAT PATTERN, FF-H02-3

SCALE 1:4. Blank 978.1 x 187.7, seen from the inside; u along, w across



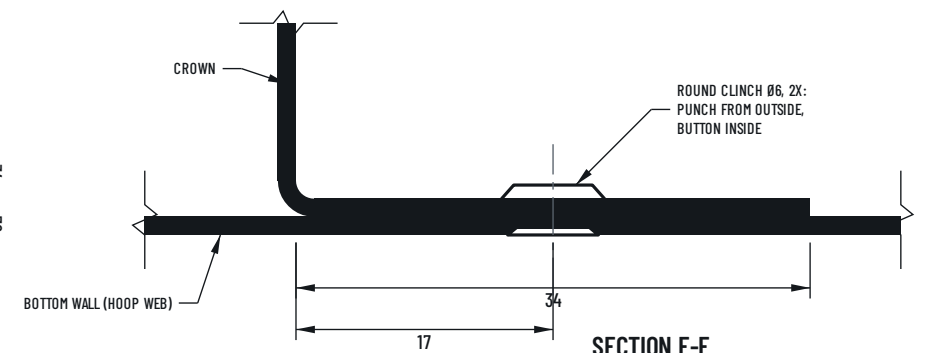
PART NO.	BLOCK HEIGHT	LEG LENGTH	OVER TABS	J GROOVES	HUB HOLES	BLANK	WHERE
FF-H02-3	3 ft	866.4	912	12, 6 per leg	None	978.1 x 187.7	3x3, seam 2 ft in
FF-H02-3H	3 ft	866.4	912	12, 6 per leg	3, right leg	978.1 x 187.7	3x3 and 3x2, 1 ft in
FF-H02-2	2 ft	561.6	607.2	8, 4 per leg	None	673.3 x 187.7	2x3, seam 2 ft in
FF-H02-2H	2 ft	561.6	607.2	8, 4 per leg	3, right leg	673.3 x 187.7	2x3 and 2x2, 1 ft in

NOTES

1. Plain channel, 12 steel strip, same coil as the hoop (sheet H-1). Both legs are near legs for the panels either side of the seam, so both carry the J grooves (± 0.1 to the lower tab's outer face).
2. At both ends the crown runs on past the legs as a tongue and folds back 90° into a tab on the wall's inner face. Clinch before coating; the outer face keeps a shallow dish, nothing proud of the joint face.
3. Second moment of area 138,397 mm⁴. At the design wind load it carries 25 MPa (9% of yield) and deflects 0.24 [0.009 in].
4. $u = \text{block } y + 31.84$; the pockets are at u 91.84, 266.64, 396.64, 571.44, 701.44, 876.24.
5. The H rails go in the seam 1 ft in from the left, seen from the front, and carry the hub: 2 x $\phi 5.1$ at y 377.2 and 537.2, d 50, and $\phi 6.5$ at y 557.2, d 42, in the right leg, with two FF-C06 pins riveted in before the rail is clinched (sheet H-5).

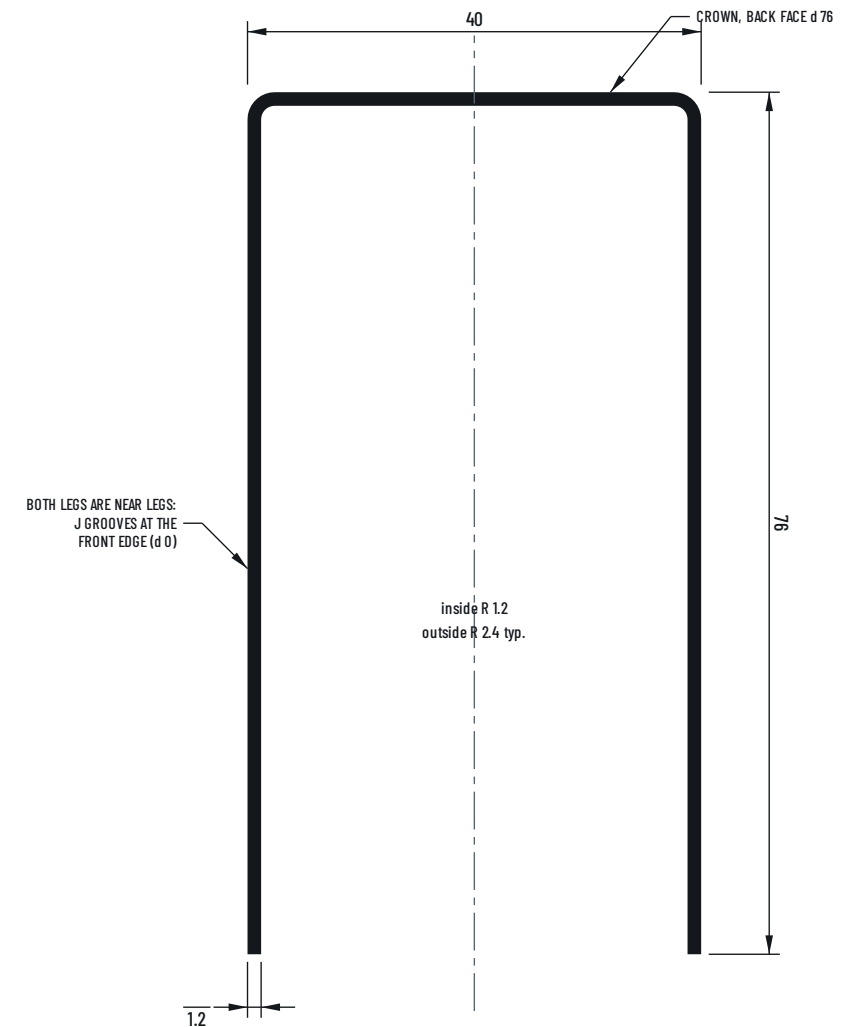
DETAIL J, TAB ON THE WALL

SCALE 2:1. Looking along x; top end alike



SECTION E-E

SCALE 1.5:1. Front edge down

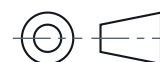


Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

FRAME: HOOP



THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

HOOP SEAM RAIL

FF-H02-3 shown; FF-H02-2 and the H rails per the table

PART NO.

FF-H02

MATERIAL
S280GD+ZM310 steel strip, 1.2 [0.047], zinc magnesium coated

FINISH
Coated with the hoop after clinching (sheet H-0)

SCALE

AS SHOWN

	UNITS
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mm

SHEET

13 of 20

DATE _____

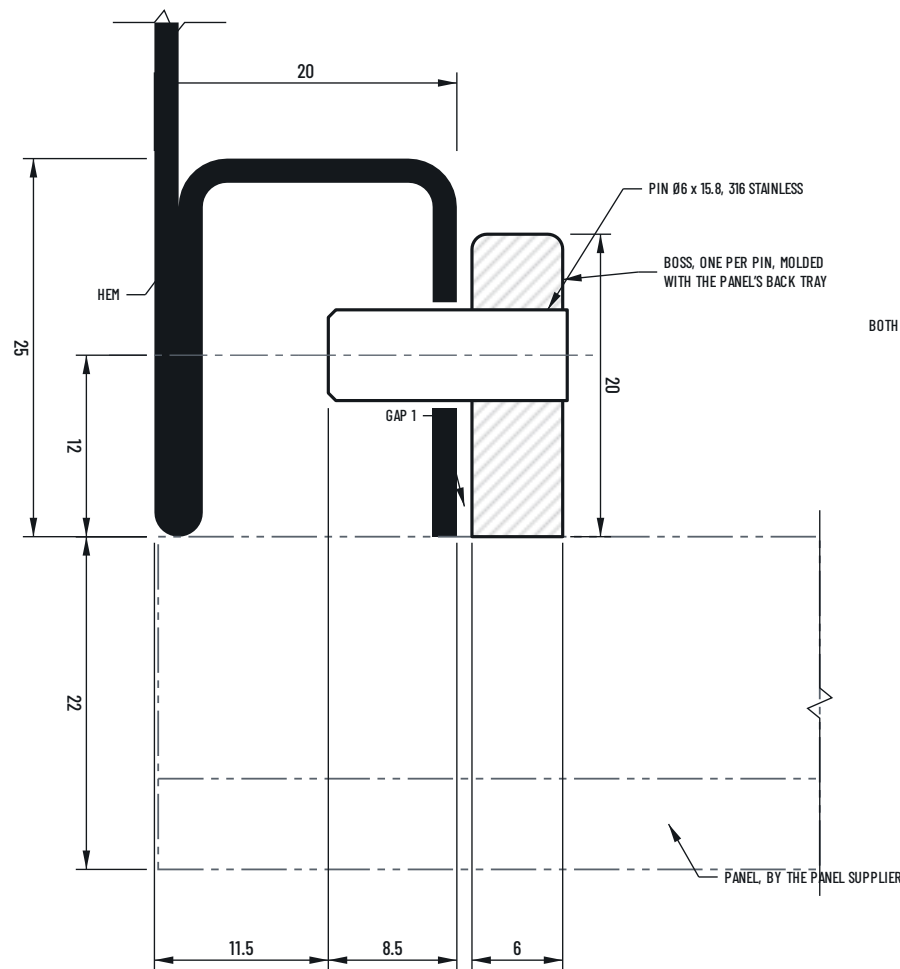
2026-10-05

-4

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05

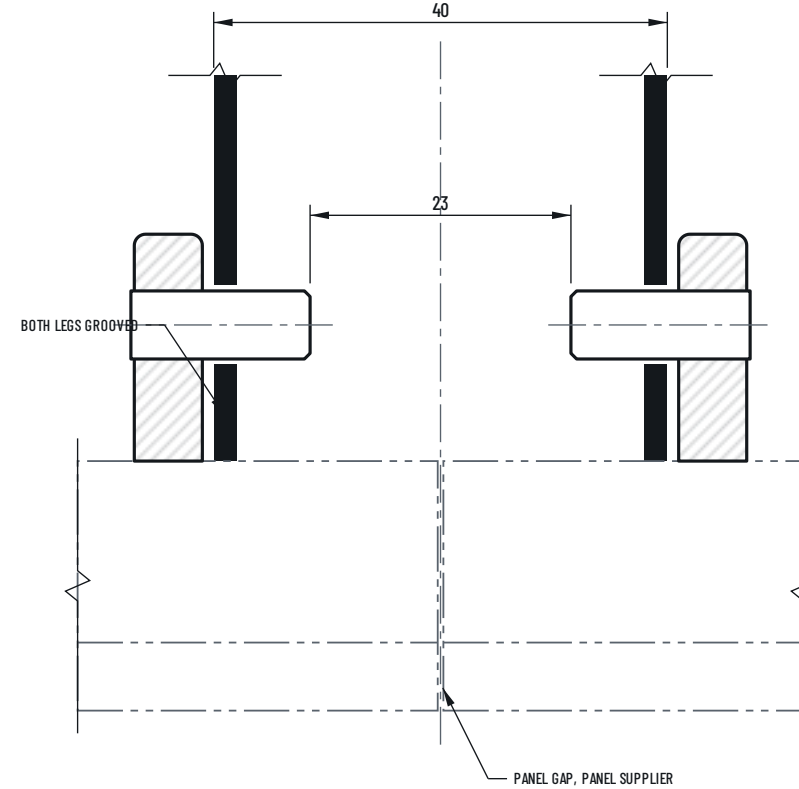
SECTION A-A, AT A SIDE WALL

SCALE 2:1. Rib shown; Hoop the same in 1.2 sheet. Front edge down



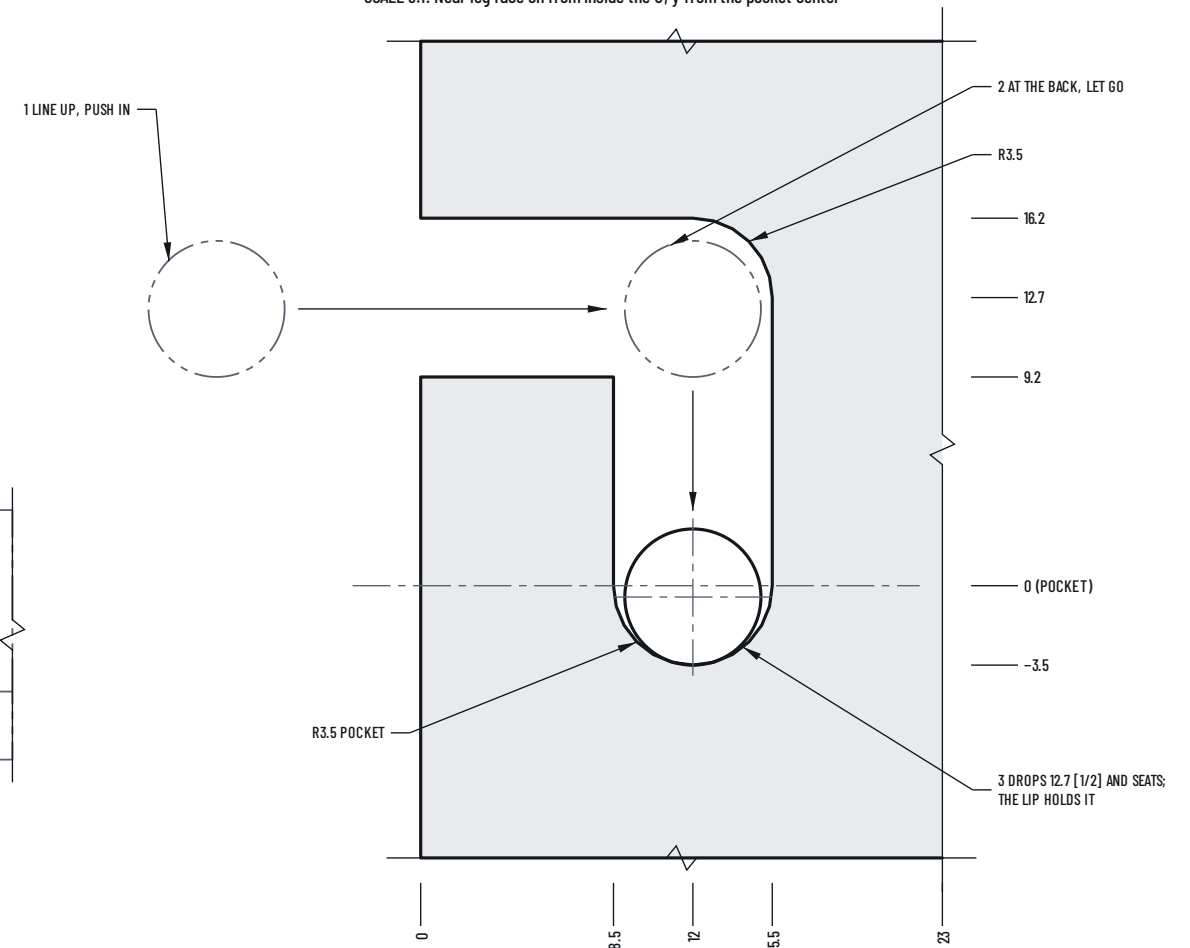
SECTION AT A SEAM, SAME PLANE

SCALE 1.5:1. Rib seam rail; its crown is at d 66, past the break



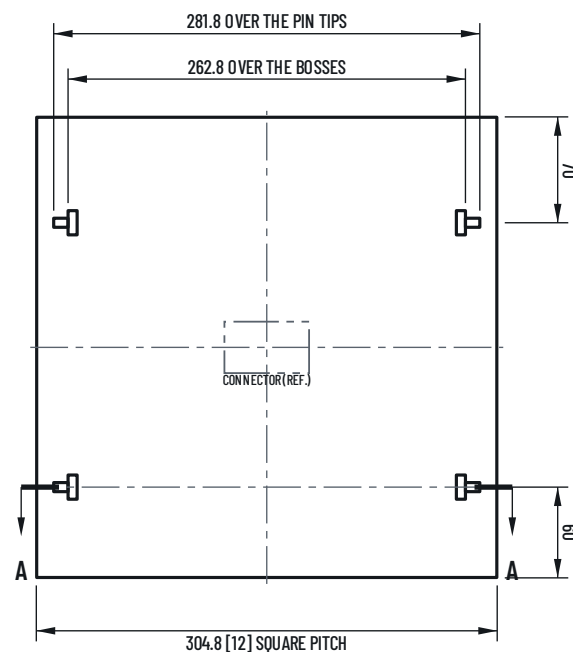
DETAIL J, THE J GROOVE

SCALE 3:1. Near leg face on from inside the U; y from the pocket center



VIEW P, PANEL BACK

SCALE 1:5. Four pins, two per side, all pointing out



POCKET CENTERS, BLOCK y FROM ITS BOTTOM FACE

HEIGHT	POCKETS; EACH ENTRY IS 12.7 ABOVE ITS POCKET
2 ft	60, 234.8, 364.8, 539.6
3 ft	60, 234.8, 364.8, 539.6, 669.6, 844.4

EVERY GROOVED LEG

PART	LEG	SHEET	DRAWING
FF-R01, FF-R02	Side wall near leg	1.6	R-1, R-2
FF-R05	Seam rail, both legs	2.0	R-4
FF-H01	Strip, both side near legs	1.2	H-2, H-3
FF-H02	Seam rail, both legs	1.2	H-4

1. Every near legs inner face sits 20.0 ± 0.15 from the wall's outer face or the seam center, on both frames.
2. Grooves are cut in the flat: ± 0.1 on the groove, ± 0.2 on the pocket centers from the parts datum end. Top and bottom walls carry no grooves.

HOW IT WORKS

1. Hang: line the four pins up with the entries, push straight back 12, let go. The panel drops 12.7 [1/2] and every pin seats in its pocket. No tools and no latch.
2. Remove: lift 12.7 first, then pull straight out. A column comes out from the top down; the panels above lift out in seconds.
3. The panel's back bears on the rails' front edges (d 0); the pins locate and hold. Wind suction, about 100 N a pin at the design load, bears on the lip; the boss stays under 10 MPa.
4. Panel side, by the panel supplier: pin $\varnothing 6$ h9 x 15.8, 316 stainless, tip chamfer $0.5 \times 45^\circ$, center 12.0 ± 0.15 behind the panel back; boss 6 thick, 16 tall, round ended.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

COMMON: RIB AND HOOP



DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

GRAVITY LOCK, PANEL TO FRAME

The J groove in every near leg; panel pins and bosses for reference

PART NO.

J GROOVE, ALL NEAR LEGS

QTY PER BLOCK

4 per panel

	MATERIAL
1	100
2	100
3	100
4	100
5	100
6	100
7	100
8	100
9	100
10	100
11	100
12	100
13	100
14	100
15	100
16	100
17	100
18	100
19	100
20	100
21	100
22	100
23	100
24	100
25	100
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83	100
84	100
85	100
86	100
87	100
88	100
89	100
90	100
91	100
92	100
93	100
94	100
95	100
96	100
97	100
98	100
99	100
100	100

MATERIAL	As the grooved part
FINISH	As the grooved part

SCALE

AS SHOWN

	UNITS
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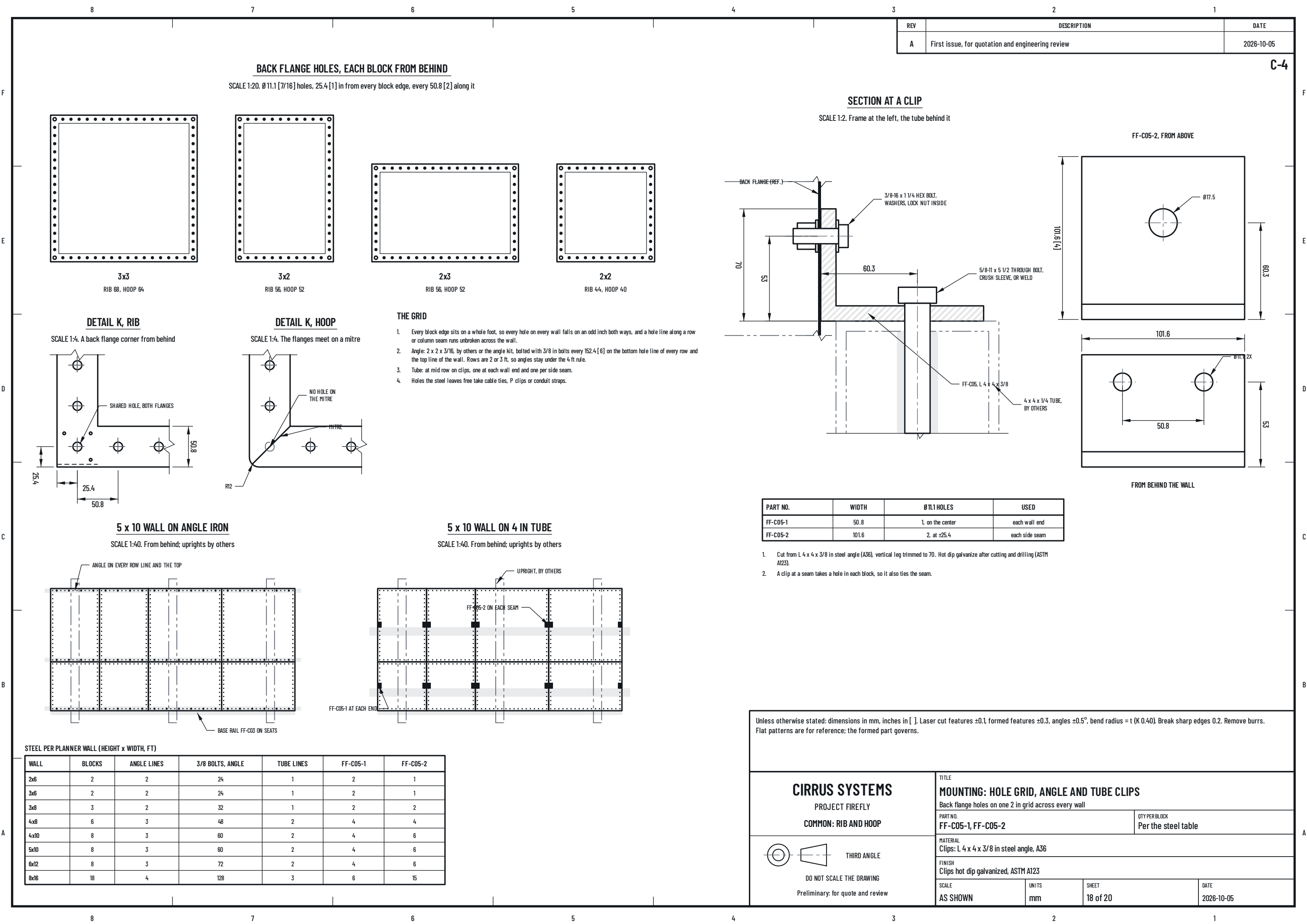
mm

	SHEET
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DATE _____

2026-10-05

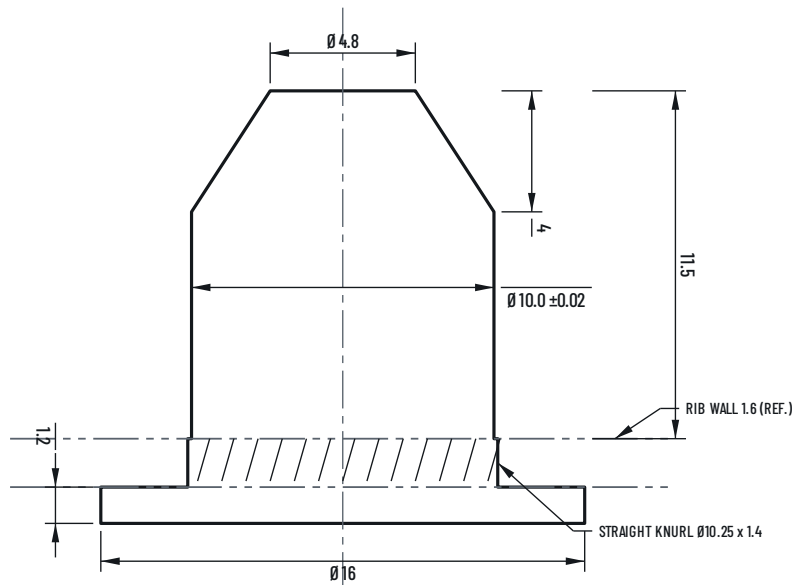


REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05

C-5

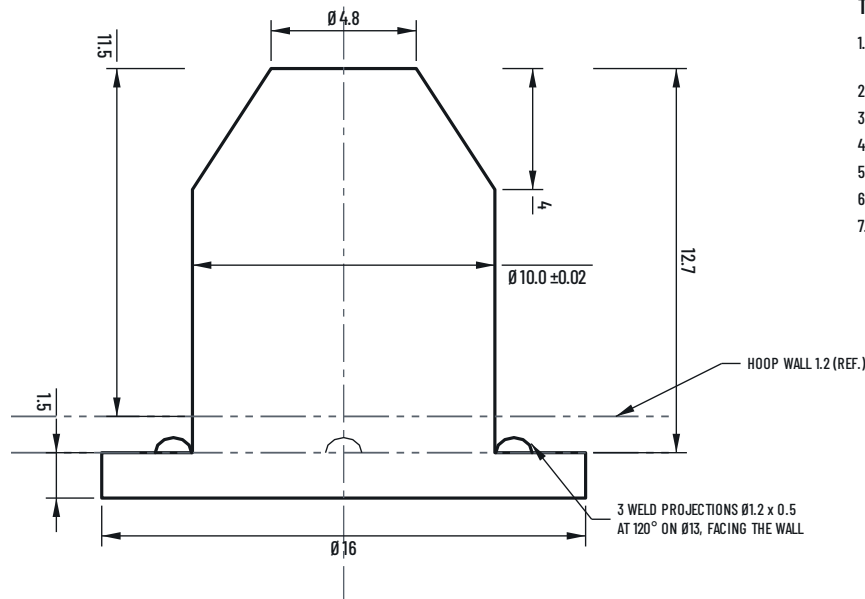
FF-C01 LOCATING PIN, PRESS FIT

SCALE 4:1. Rib. 303 stainless



FF-C02 LOCATING PIN, PROJECTION WELD

SCALE 4:1. Hoop. Low carbon steel, zinc nickel 8 μ m

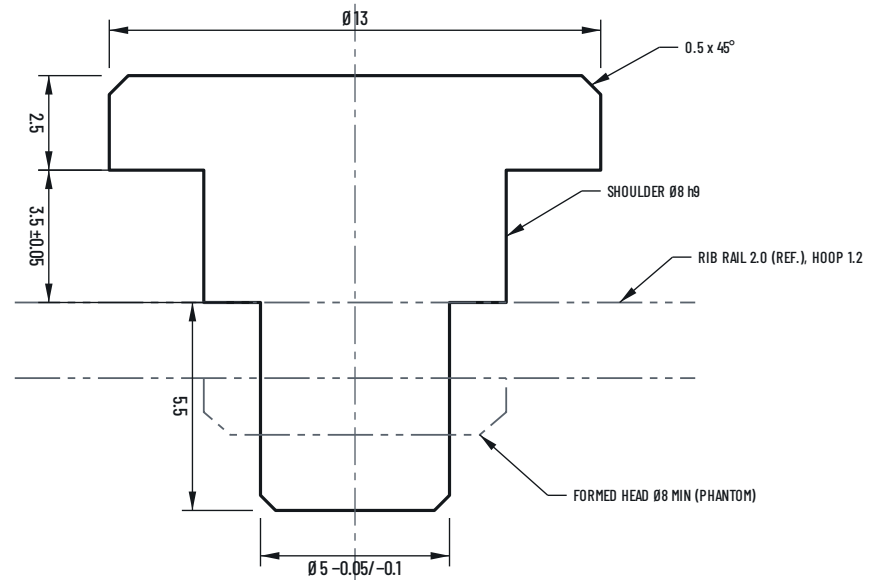


THE PINS

- Both pins stand 11.5 above the top walls outer face, $\varnothing 10.0$ with a 4 mm taper to $\varnothing 4.8$, and enter $\varnothing 10.2 \pm 0.1/0$ sockets in the bottom wall of the block above. The taper finds the socket before the walls meet.
- Only the lowest 1.6 of the pin locates. Position: ± 0.05 pin to pin on a block, ± 0.1 to the frame datums (sheets R-3, H-3). A block lands within about 0.25 of where it belongs.
- FF-C01 presses from inside into $\varnothing 10.0 \pm 0.05/0$ until the collar seats. Push out at least 1 kN, checked on first articles.
- FF-C02 projection welds from inside, through $\varnothing 10.2 \pm 0.1/0$, after the ring is closed. Weld before powder coat.
- Mask both pins, the sockets and every nut thread during powder coat.
- FF-C04, the rail pin, is on sheet C-3.
- FF-C06, the hub pin, is orbital riveted from inside the H seam rail before coating (sheets R-5, H-5): shoulder flat on the rail within 0.1, formed head $\varnothing 8$ min. Its 3.5 shoulder leaves 1.5 of play for the hubs 2.0 hook plate.

FF-C06 HUB PIN, ORBITAL RIVET

SCALE 5:1. Rib and Hoop. 316 stainless, passivated



HARDWARE, PER BLOCK (EACH BLOCK SIZE, HEIGHT x WIDTH, FT)

PART	DESCRIPTION	MATERIAL	WHERE	3x3	3x2	2x3	2x2	3x3	3x2	2x3	2x2
ISO 4162	Hex flange bolt M8 x 16, class 8.8	Zinc flake	Every joint, ships loose	7	6	5	4	7	6	5	4
CLS-M8-1	PEM self clinching nut M8	303 stainless	Rib top and right side walls	7	6	5	4				
DIN 929 M8	Hex weld nut M8	Steel, plain	Hoop top and right side walls					7	6	5	4
FF-C01	Locating pin, press fit	303 stainless	Rib top wall	3	2	3	2				
FF-C02	Locating pin, projection weld	Steel, zinc nickel	Hoop top wall					3	2	3	2
FF-C06	Hub pin, orbital riveted	316 stainless	Seam rail at the hub (R-5, H-5)	2	2	2	2	2	2	2	2
RIV-32-CSK	Blind rivet Ø 3.2, closed end, 100° head	Al body, A2 mandrel	Rib corners, tabs and seam rails	32	28	32	28				

HARDWARE, PER WALL

PART	DESCRIPTION	MATERIAL	QUANTITY	SHEET
FF-C03-n	Base rail section, 6 x 2 x 1/4 in tube, n feet	6061-T6	1 per 10 ft of wall or less	C-3
FF-C04	Rail pin, threaded, fitted to the rail	303 stainless	1 per foot of wall	C-3
ISO 4162	Hex flange bolt M10 x 20, class 8.8, rail to seat	Zinc flake	2 per seat	C-3
CAP	End cap for 6 x 2 x 1/4 in tube	Black LDPE	2 per wall	C-3
FF-C05-1, -2	Tube clip, single and double	A36, galvanized	Per the steel table	C-4
CLIP BOLTS	3/8-16 x 1 1/4 hex bolt, 2 washers, nylon lock nut	Zinc plated, grade 5	1 per Ø1.1 clip hole	C-4
TUBE BOLTS	5/8-11 x 5 1/2 hex bolt, crush sleeve, washers, lock nut	Zinc plated, grade 5	1 per clip, or weld	C-4

WHAT SHIPS WITH WHAT

1. Each block ships with its own joint bolts: one for every clearance hole in its bottom wall and its left side wall (seen from the front). The bolts go into the next block's nuts, or into the base rail.
2. The base rail kit ships with its pins fitted, its M10 bolts and two end caps. Clips ship with their bolts. Angle iron and its bolts, the seats and the uprights are by others, or the optional angle kit.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

COMMON: RIB AND HOOP



DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

FRAME PINS AND HARDWARE

The three frame pins and every purchased part, per block and per wall

PART NO.

FF-C01, FF-C02, FF-C06

QTY PER BLOCK

Per the tables

[illegible]

As listed

FINISH

SCALE

AS SHOWN

UNITS

mm

SHEET

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DATE _____

2026-10-05

